

Database System Engineering and Distributed Backend Development

JOINS AND SET OPERATIONS

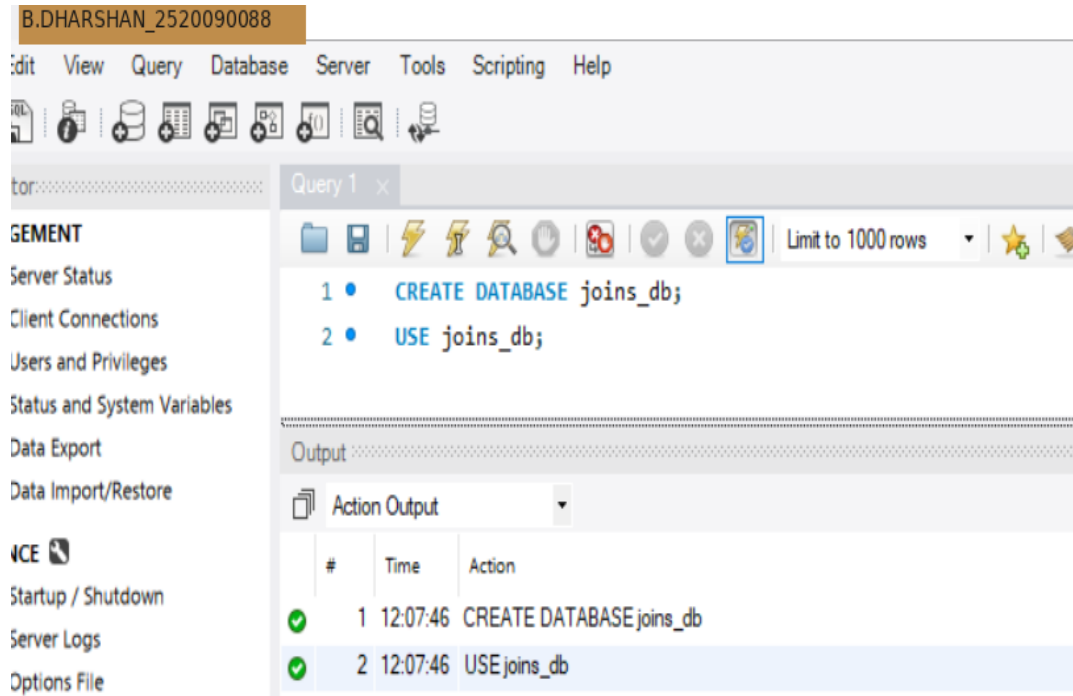
Name: B.DHARSHAN

Roll No: 2520090088

Section: 11

Step 1 — Create and Select a Database

```
CREATE DATABASE joins_db;  
USE joins_db;
```

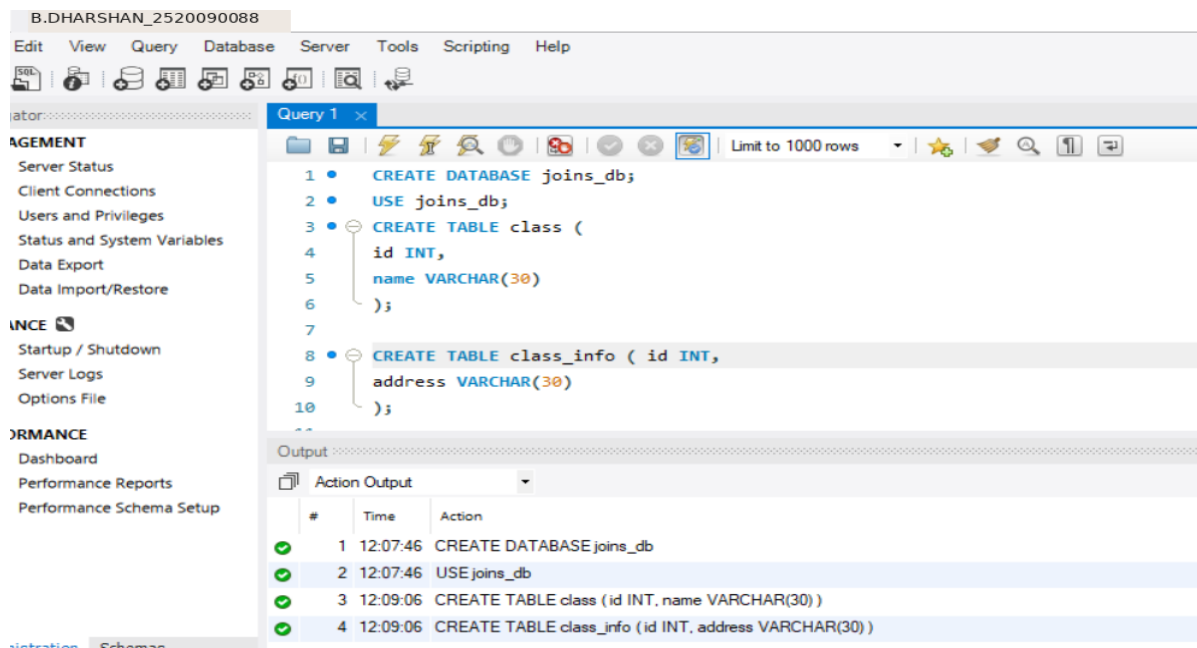


Query:

Step 2 — Create the Tables (class, class_info)

Query:

```
CREATE TABLE class (  
    id INT,  
    name VARCHAR(30)  
);  
  
CREATE TABLE class_info (  
    id INT,  
    address VARCHAR(30)  
);
```



Step 3 — Insert Data

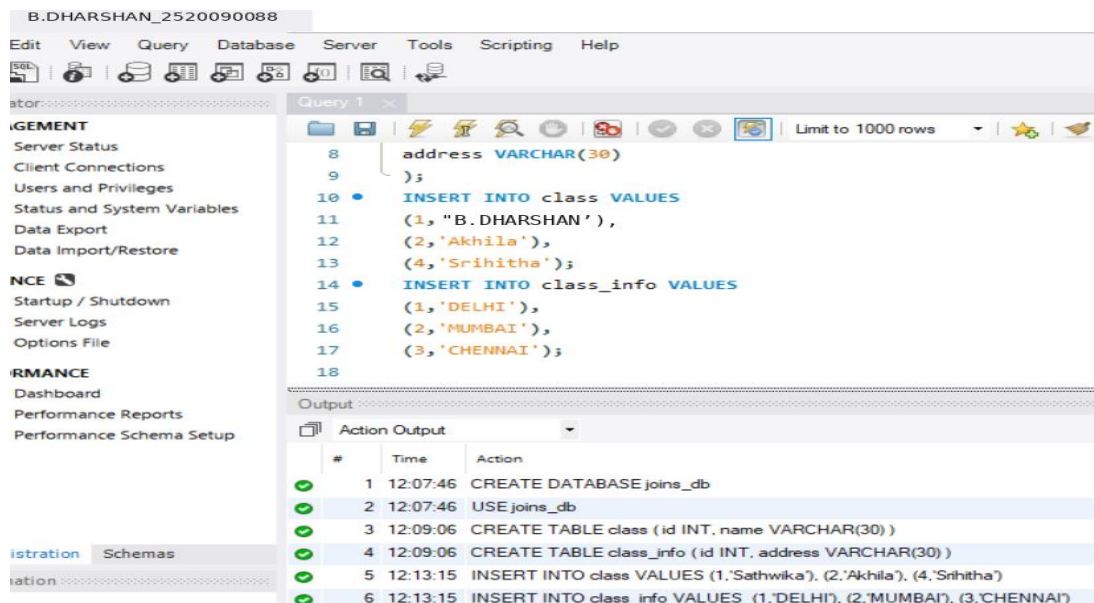
Query:

```

INSERT INTO class VALUES
(1,'B.DHARSHAN'),
(2,'Akhila'),
(4,'Srihitha');

INSERT INTO class_info VALUES
(1,'DELHI'),
(2,'MUMBAI'),
(3,'CHENNAI');

```



Step 4 — Question 1: CROSS JOIN

Query:

```
SELECT *  
FROM class  
CROSS JOIN class_info;
```

The screenshot shows a database management tool interface. The top menu bar includes 'Edit', 'View', 'Query', 'Database', 'Server', 'Tools', 'Scripting', and 'Help'. The left sidebar contains a tree view with categories like 'iGEMENT', 'NCE', and 'RMANCE'. The main window displays 'Query 1' with the following SQL code:

```
16 (2, 'MUMBAI'),  
17 (3, 'CHENNAI');  
18 • SELECT *  
19 FROM class  
20 CROSS JOIN class_info;  
21  
22
```

Below the query editor, the 'Result Grid' shows the output of the query. It is a table with 4 columns: 'id', 'name', 'id', and 'address'. The data is as follows:

	id	name	id	address
▶	4	Srihitha	1	DELHI
	2	Akhila	1	DELHI
	1	B.DHARSHAN		ELHI
	4	Srihitha	2	MUMBAI
	2	Akhila	2	MUMBAI
	1	Sathwika	2	MUMBAI
	4	Srihitha	3	CHENNAI
	2	Akhila	3	CHENNAI
	1	Sathwika	3	CHENNAI

At the bottom, the 'Output' pane shows the execution log with the following entries:

#	Time	Action
✓ 2	12:07:46	USE joins_db
✓ 3	12:09:06	CREATE TABLE class (id INT, name VARCHAR(30))
✓ 4	12:09:06	CREATE TABLE class_info (id INT, address VARCHAR(30))
✓ 5	12:13:15	INSERT INTO class VALUES (1, 'B.DHARSHAN'), (2, 'Srihitha')
✓ 6	12:13:15	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
✓ 7	12:15:12	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000

Step 5 — Recreate Tables for INNER JOIN Section

Query:

```
DROP TABLE class, class_info;  
  
CREATE TABLE class (  
    id INT,  
    name VARCHAR(30)  
);  
  
CREATE TABLE class_info (  
    id INT,  
    address VARCHAR(30)  
);
```

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Query 1

```

16 (2,'MUMBAI'),
17 (3,'CHENNAI');
18 • SELECT *
19 FROM class
20 CROSS JOIN class_info;
21 • DROP TABLE class, class_info;
22
23 • CREATE TABLE class (
24     id INT,
25     name VARCHAR(30)
26 );
27
28 • CREATE TABLE class_info ( id INT,
29     address VARCHAR(30)
30 );
31
32

```

Output

Action Output

#	Time	Action
2	12:07:46	USE joins_db
3	12:09:06	CREATE TABLE class (id INT, name VARCHAR(30))
4	12:09:06	CREATE TABLE class_info (id INT, address VARCHAR(30))
5	12:13:15	INSERT INTO class VALUES (1, 'B.DHARSHAN'), (1, 'Srihitha')
6	12:13:15	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
7	12:15:12	SELECT * FROM class CROSS JOIN class_info
8	12:16:28	DROP TABLE class, class_info
9	12:16:28	CREATE TABLE class (id INT, name VARCHAR(30))
10	12:16:28	CREATE TABLE class_info (id INT, address VARCHAR(30))

Step 6 — Insert Updated Data

Query:

```

INSERT INTO class VALUES
(1,'B.DHARSHAN'),
(2,'Nikki'),
(3,'Akhila'),
(4,'Srihitha');

INSERT INTO class_info VALUES
(1,'DELHI'),
(2,'MUMBAI'),
(3,'CHENNAI');

```

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Query 1

```

29 address VARCHAR(30)
30 );
31 • INSERT INTO class VALUES
32 (1, 'B. DHARSHAN'),
33 (2, 'Nikki'),
34 (3, 'Akhila'),
35 (4, 'Srihitha');
36
37 • INSERT INTO class_info VALUES
38 (1, 'DELHI'),
39 (2, 'MUMBAI'),
40 (3, 'CHENNAI');
41

```

Output

Action Output

#	Time	Action
1	12:07:46	CREATE DATABASE joins_db
2	12:07:46	USE joins_db
3	12:09:06	CREATE TABLE class (id INT, name VARCHAR(30))
4	12:09:06	CREATE TABLE class_info (id INT, address VARCHAR(30))
5	12:13:15	INSERT INTO class VALUES (1, B. DHARSHAN),
6	12:13:15	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
7	12:15:12	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000
8	12:16:28	DROP TABLE class, class_info
9	12:16:28	CREATE TABLE class (id INT, name VARCHAR(30))
10	12:16:28	CREATE TABLE class_info (id INT, address VARCHAR(30))
11	12:18:13	INSERT INTO class VALUES (1, B. DHARSHAN), (2, 'Nikki'), (3, 'Akhila'), (4, 'Srihitha')
12	12:18:13	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')

Step 7 — Question 2: INNER JOIN

Query:

```

SELECT *
FROM class
INNER JOIN class_info
ON class.id = class_info.id;

```

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Query 1

```

42 FROM class
43 INNER JOIN class_info
44 ON class.id = class_info.id;
45
46

```

Result Grid

	id	name	id	address
1	1	Sathwika	1	DELHI
2	2	Nikki	2	MUMBAI
3	3	Akhila	3	CHENNAI

Result 2

Output

Action Output

#	Time	Action	Message
1	12:07:46	CREATE DATABASE joins_db	1 row(s)
2	12:07:46	USE joins_db	0 row(s)
3	12:09:06	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s)
4	12:09:06	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s)
5	12:13:15	INSERT INTO class VALUES (1, 'B. DHARSHAN'), (2, 'Nikki'), (3, 'Akhila'), (4, 'Srihitha')	3 row(s)
6	12:13:15	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')	3 row(s)
7	12:15:12	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s)
8	12:16:28	DROP TABLE class, class_info	0 row(s)
9	12:16:28	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s)
10	12:16:28	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s)
11	12:18:13	INSERT INTO class VALUES (1, 'B. DHARSHAN'), (2, 'Nikki'), (3, 'Akhila'), (4, 'Srihitha')	4 row(s)
12	12:18:13	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')	3 row(s)
13	12:26:56	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	3 row(s)

Step 8 — Question 3: INNER JOIN (Selected Columns)

Query:

```
SELECT class.name,  
       class_info.address  
FROM class  
INNER JOIN class_info  
ON class.id = class_info.id;
```

The screenshot shows the MySQL Workbench interface. The top toolbar includes icons for File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The left sidebar contains a tree view with categories like 'Environment', 'Server Status', 'Client Connections', 'Users and Privileges', 'Status and System Variables', 'Data Export', 'Data Import/Restore', 'Performance', 'Performance Reports', and 'Performance Schema Setup'. The main window displays 'Query 1' with the following SQL code:

```
45 ON class.id = class_info.id;  
46 SELECT class.name,  
47 class_info.address FROM class  
48 INNER JOIN class_info  
49 ON class.id = class_info.id;
```

Below the query editor, the 'Result Grid' shows the output of the query. It has columns 'name' and 'address' and contains three rows:

name	address
B. DHARSHAN	
Nikki	MUMBAI
Akhila	CHENNAI

The bottom pane shows the 'Action Output' log, which lists 15 actions performed, including creating the database 'joins_db', creating tables 'class' and 'class_info', inserting data, and executing the INNER JOIN query.

Step 9 — Question 4: NATURAL JOIN

Query:

```
SELECT *  
FROM class  
NATURAL JOIN class_info;
```

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Query 1 x

```

49 ON class.id = class_info.id;
50 SELECT *
51 FROM class
52 NATURAL JOIN class_info;
53

```

Result Grid

	id	name	address
▶	1	B. DHARSHAN	
	2	Nikki	MUMBAI
	3	Akhila	CHENNAI

Result 6 x

Output

Action Output

#	Time	Action
3	12:29:46	USE joins_db
4	12:29:46	CREATE TABLE class (id INT, name VARCHAR(30))
5	12:29:46	CREATE TABLE class_info (id INT, address VARCHAR(30))
6	12:29:46	INSERT INTO class VALUES (1, 'B. DHARSHAN'), (1, 'Srihitha')
7	12:29:46	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
8	12:29:46	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000
9	12:29:46	DROP TABLE class, class_info
10	12:29:46	CREATE TABLE class (id INT, name VARCHAR(30))
11	12:29:46	CREATE TABLE class_info (id INT, address VARCHAR(30))
12	12:29:46	INSERT INTO class VALUES (1, 'B. DHARSHAN'), (4, 'Srihitha')
13	12:29:46	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
14	12:29:46	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
15	12:29:46	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
16	12:30:54	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000

Step 10 — Insert Extra Rows for Outer Join Section

Query:

```

INSERT INTO class VALUES
(5, 'Rohan');

```

```

INSERT INTO class_info VALUES
(7, 'NOIDA'),
(8, 'PANIPAT');

```

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Query 1 x

```

49 ON class.id = class_info.id;
50 SELECT *
51 FROM class
52 NATURAL JOIN class_info;
53 INSERT INTO class VALUES
54 (5, 'Rohan');
55
56 INSERT INTO class_info VALUES
57 (7, 'NOIDA'),
58 (8, 'PANIPAT');
59
60

```

Output

Action Output

#	Time	Action
5	12:29:46	CREATE TABLE class_info (id INT, address VARCHAR(30))
6	12:29:46	INSERT INTO class VALUES (1, 'B. DHARSHAN'), (1, 'Srihitha')
7	12:29:46	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
8	12:29:46	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000
9	12:29:46	DROP TABLE class, class_info
10	12:29:46	CREATE TABLE class (id INT, name VARCHAR(30))
11	12:29:46	CREATE TABLE class_info (id INT, address VARCHAR(30))
12	12:29:46	INSERT INTO class VALUES (1, 'B. DHARSHAN'), (4, 'Srihitha')
13	12:29:46	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
14	12:29:46	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
15	12:29:46	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
16	12:30:54	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000
17	12:32:40	INSERT INTO class VALUES (5, 'Rohan')
18	12:32:40	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')

Step 11 — Question 5: LEFT OUTER JOIN

Query:

```
SELECT *
FROM class
LEFT OUTER JOIN class_info
ON class.id = class_info.id;
```

The screenshot shows a database management tool interface. The top menu bar includes 'Edit', 'View', 'Query', 'Database', 'Server', 'Tools', 'Scripting', and 'Help'. The left sidebar contains a tree view with categories like 'GEMENT', 'VCE', and 'RMANCE'. The main area displays 'Query 1' with the following SQL code:

```
58 (8, 'PANIPAT');
59 SELECT *
60 FROM class
61 LEFT OUTER JOIN class_info ON class.id = class_info.id;
62
63
```

Below the query, the 'Result Grid' shows the following data:

	id	name	id	address
1	1	Sathwika	1	DELHI
2	2	Nikki	2	MUMBAI
3	3	Akhila	3	CHENNAI
4	4	Srihitha	NULL	NULL
5	5	Rohan	NULL	NULL

Below the result grid, the 'Action Output' section shows a list of actions performed, including creating tables, inserting data, and executing the query. The actions are as follows:

#	Time	Action
8	12:29:46	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000
9	12:29:46	DROP TABLE class, class_info
10	12:29:46	CREATE TABLE class (id INT, name VARCHAR(30))
11	12:29:46	CREATE TABLE class_info (id INT, address VARCHAR(30))
12	12:29:46	INSERT INTO class VALUES (1, 'B. DHARSHAN'), (4, 'Srihitha')
13	12:29:46	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
14	12:29:46	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
15	12:29:46	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id
16	12:30:54	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000
17	12:32:40	INSERT INTO class VALUES (5, 'Rohan')
18	12:32:40	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')
19	12:34:13	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000

Step 12 — Question 6: LEFT JOIN (Unmatched Only)

Query:

```
SELECT *
FROM class
LEFT JOIN class_info
ON class.id = class_info.id
WHERE class_info.id IS NULL;
```

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Query 1 x

```

60 FROM class
61 LEFT OUTER JOIN class_info ON class.id = class_info.id;
62 SELECT *
63 FROM class
64 LEFT JOIN class_info
65 ON class.id = class_info.id WHERE class_info.id IS NULL;

```

id	name	id	address
4	Srihitha	NULL	NULL
5	Rohan	NULL	NULL

Result 8 x

Output

Action Output

#	Time	Action
9	12:29:46	DROP TABLE class, class_info
10	12:29:46	CREATE TABLE class (id INT, name VARCHAR(30))
11	12:29:46	CREATE TABLE class_info (id INT, address VARCHAR(30))
12	12:29:46	INSERT INTO class VALUES (1, 'B. DHARSHAN'), (4, 'Srihitha')
13	12:29:46	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
14	12:29:46	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
15	12:29:46	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
16	12:30:54	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000
17	12:32:40	INSERT INTO class VALUES (5, 'Rohan')
18	12:32:40	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')
19	12:34:13	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
20	12:35:32	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000

Info Session

Step 13 — Question 7: RIGHT OUTER JOIN

Query:

```

SELECT *
FROM class
RIGHT OUTER JOIN class_info
ON class.id = class_info.id;

```

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Edit View Query Database Server Tools Scripting Help

Query 1 x

```

60 FROM class
61 LEFT OUTER JOIN class_info ON class.id = class_info.id;
62 SELECT *
63 FROM class
64 LEFT JOIN class_info
65 ON class.id = class_info.id WHERE class_info.id IS NULL;

```

id	name	id	address
4	Srihitha	NULL	NULL
5	Rohan	NULL	NULL

Result 9 Result 10 Result 11 Result 12 Result 13 Result 14 x

Output

Action Output

#	Time	Action
9	12:39:32	DROP TABLE class, class_info
10	12:39:32	CREATE TABLE class (id INT, name VARCHAR(30))
11	12:39:32	CREATE TABLE class_info (id INT, address VARCHAR(30))
12	12:39:32	INSERT INTO class VALUES (1, 'B. DHARSHAN'), (4, 'Srihitha')
13	12:39:32	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
14	12:39:32	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
15	12:39:32	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
16	12:39:32	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000
17	12:39:32	INSERT INTO class VALUES (5, 'Rohan')
18	12:39:32	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')
19	12:39:32	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
20	12:39:32	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000

Info Session

Step 14 — Question 8: RIGHT JOIN (Unmatched Only)

Query:

```
SELECT *
FROM class
RIGHT JOIN class_info
ON class.id = class_info.id
WHERE class.id IS NULL;
```

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree with 'Server Status', 'Client Connections', 'Users and Privileges', 'Status and System Variables', 'Data Export', and 'Data Import/Restore' under 'SERVER'. The 'Performance' section includes 'Dashboard', 'Performance Reports', and 'Performance Schema Setup'. The 'Schemas' section is also visible. The right pane shows 'Query 1' with the following SQL query:

```
64 LEFT JOIN class_info
65 ON class.id = class_info.id WHERE class_info.id IS NULL;
66 SELECT *
67 FROM class
68 RIGHT JOIN class_info
69 ON class.id = class_info.id WHERE class.id IS NULL;
```

Below the query, the 'Result Grid' shows the results of the query. The columns are 'id', 'name', 'id', and 'address'. The results are:

id	name	id	address
7		7	NOIDA
8		8	PANIPAT

The 'Output' pane shows the execution plan for the query. The actions are:

- 10 12:39:32 CREATE TABLE class (id INT, name VARCHAR(30))
- 11 12:39:32 CREATE TABLE class_info (id INT, address VARCHAR(30))
- 12 12:39:32 INSERT INTO class VALUES (1, 'B. DHARSHAN'), (2, 'Srihitha')
- 13 12:39:32 INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHENNAI')
- 14 12:39:32 SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
- 15 12:39:32 SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
- 16 12:39:32 SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000
- 17 12:39:32 INSERT INTO class VALUES (5, 'Rohan')
- 18 12:39:32 INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')
- 19 12:39:32 SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
- 20 12:39:32 SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL
- 21 12:41:16 SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL

Step 15 — Question 9: FULL OUTER JOIN

Query:

```
SELECT *
FROM class
LEFT JOIN class_info
ON class.id = class_info.id
UNION
SELECT *
FROM class
RIGHT JOIN class_info
ON class.id = class_info.id;
```

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Query 1

```

70 SELECT *
71 FROM class
72 LEFT JOIN class_info
73 ON class.id = class_info.id UNION
74 SELECT *
75 FROM class
76 RIGHT JOIN class_info
77 ON class.id = class_info.id;
78

```

id	name	id	address
1	B.DHARSHAN	ELHI	
2	Nikdo	2	MUMBAI
3	Akhila	3	CHENNAI
4	Srihitha	NULL	NULL
5	Rohan	NULL	NULL
7	NOIDA		
8	PANIPAT		

Result 16 Result 17 Result 18 Result 19 Result 20 Result 21 Result 22 Result 23

Output

Action Output

#	Time	Action
16	12:42:05	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000
17	12:42:05	INSERT INTO class VALUES (5,'Rohan')
18	12:42:05	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')
19	12:42:05	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
20	12:42:05	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL
21	12:42:05	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL
22	12:42:05	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id UNION SELECT * FROM class

Step 16 – Question 10: FULL OUTER JOIN (Unmatched Only)

Query:

```

SELECT *
FROM class
LEFT JOIN class_info
ON class.id = class_info.id
WHERE class_info.id IS NULL
UNION
SELECT *
FROM class
RIGHT JOIN class_info
ON class.id = class_info.id
WHERE class.id IS NULL;

```

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Edit View Query Database Server Tools Scripting Help

Query 1

```

78 SELECT *
79 FROM class
80 LEFT JOIN class_info
81 ON class.id = class_info.id WHERE class_info.id IS NULL UNION
82 SELECT *
83 FROM class
84 RIGHT JOIN class_info
85 ON class.id = class_info.id WHERE class.id IS NULL;
86

```

id	name	id	address
4	Srihitha	NULL	NULL
5	Rohan	NULL	NULL
7	NOIDA		
8	PANIPAT		

Result 24 Result 25 Result 26 Result 27 Result 28 Result 29 Result 30

Output

Action Output

#	Time	Action
17	12:43:16	INSERT INTO class VALUES (5,'Rohan')
18	12:43:16	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')
19	12:43:16	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
20	12:43:16	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL
21	12:43:16	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL
22	12:43:16	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id UNION SELECT * FROM class
23	12:43:16	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL

Step 17 — Create first_table and second_table (for UNION section)

Query:

```
CREATE TABLE first_table(  
  id INT,  
  name VARCHAR(30)  
);  
  
CREATE TABLE second_table(  
  id INT,  
  name VARCHAR(30)  
);  
  
INSERT INTO first_table VALUES  
(1,'Hansika'),  
(2,'Hasini');  
  
INSERT INTO second_table VALUES  
(2,'Hasini'),  
(3,'Rohan');
```

The screenshot displays a database management tool interface. The top menu bar includes 'File', 'Edit', 'Query', 'Database', 'Server', 'Tools', 'Scripting', and 'Help'. The left sidebar contains a tree view with categories like 'GENERAL', 'SCHEMAS', and 'OBJECTS'. The main window shows 'Query 1' with the following SQL script:

```
94 • INSERT INTO first_table VALUES (1,'Hansika'),  
95   (2,'Hasini');  
96  
97 • INSERT INTO second_table VALUES (2,'Hasini'),  
98   (3,'Rohan');  
99  
100  
101  
102
```

Below the query editor, the 'Result Grid' shows a table with 4 columns: 'id', 'name', 'id', and 'address'. The data is as follows:

	id	name	id	address
4	4	Srinitha	NULL	NULL
5	5	Rohan	NULL	NULL
7	NULL	NULL	7	NOIDA
8	NULL	NULL	8	PANIPAT

At the bottom, the 'Output' pane shows the execution log for the queries, including the 'CREATE TABLE' and 'INSERT INTO' statements.

Step 18 — Question 11: UNION

Query:

```
SELECT * FROM first_table
UNION
SELECT * FROM second_table;
```

The screenshot shows the MySQL Workbench interface. The top toolbar includes icons for Edit, View, Query, Database, Server, Tools, Scripting, and Help. The left sidebar contains a tree view with categories like SELEMENT, ICE, and RMANCE. The main area is divided into a Query Editor and a Results pane.

Query Editor: The query is as follows:

```
94 • INSERT INTO first_table VALUES (1,'Hansika'),
95 • (2,'Hasini');
96
97 • INSERT INTO second_table VALUES (2,'Hasini'),
98 • (3,'Rohan');
99 • SELECT * FROM first_table UNION
100 • SELECT * FROM second_table;
101 • SELECT name FROM first_table UNION
102 • SELECT name FROM second_table;
```

Results Pane: The Results pane shows the output of the query. It includes a table with the following data:

name
Hansika
Hasini
Rohan

Below the Results pane, the Action Output pane shows the execution log:

#	Time	Action
46	12:45:04	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE clas
47	12:45:04	CREATE TABLE first_table(id INT, name VARCHAR(30))
48	12:45:04	CREATE TABLE second_table(id INT, name VARCHAR(30))
49	12:45:04	INSERT INTO first_table VALUES (1,'Hansika'), (2,'Hasini')
50	12:45:04	INSERT INTO second_table VALUES (2,'Hasini'), (3,'Rohan')
51	12:46:18	SELECT * FROM first_table UNION SELECT * FROM second_table
52	12:46:18	SELECT name FROM first_table UNION SELECT name FROM second_table

Step 19 — Question 13: UNION ALL

Query:

```
SELECT * FROM first_table
UNION ALL
SELECT * FROM second_table;
```

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Edit View Query Database Server Tools Scripting Help

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GEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

NCE

- Startup / Shutdown
- Server Logs
- Options File

RMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

stration Schemas

ation.....

object selected

```

96
97 • INSERT INTO second_table VALUES (2,'Hasini'),
98 (3,'Rohan');
99 • SELECT * FROM first_table UNION
100 SELECT * FROM second_table;
101 • SELECT name FROM first_table UNION
102 SELECT name FROM second_table;
103 • SELECT * FROM first_table UNION ALL
104 SELECT * FROM second_table;

```

id	name
1	Hansika
2	Hasini
2	Hasini
3	Rohan

Result 44 x

Output

Action Output

#	Time	Action
47	12:45:04	CREATE TABLE first_table(id INT, name VARCHAR(30))
48	12:45:04	CREATE TABLE second_table(id INT, name VARCHAR(30))
49	12:45:04	INSERT INTO first_table VALUES (1,'Hansika'), (2,'Hasini')
50	12:45:04	INSERT INTO second_table VALUES (2,'Hasini'), (3,'Rohan')
51	12:46:18	SELECT * FROM first_table UNION SELECT * FROM second_table
52	12:46:18	SELECT name FROM first_table UNION SELECT name FROM second_table
53	12:48:10	SELECT * FROM first_table UNION ALL SELECT * FROM second_table

Step 21 — Question 14: UNION ALL (Row Count)

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SEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

ICE

- Startup / Shutdown
- Server Logs
- Options File

IMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

stration Schemas

ition.....

object selected

```

102 SELECT name FROM second_table;
103 • SELECT * FROM first_table UNION ALL
104 SELECT * FROM second_table;
105 • SELECT COUNT(*) FROM
106 (
107 SELECT * FROM first_table UNION ALL
108 SELECT * FROM second_table
109 ) A;
110

```

COUNT(*)
4

ult 48 Result 49 Result 50 Result 51 Result 52 Result 53 Result 54 Re

Output

Action Output

#	Time	Action
78	12:49:16	CREATE TABLE second_table(id INT, name VARCHAR(30))
79	12:49:16	INSERT INTO first_table VALUES (1,'Hansika'), (2,'Hasini')
80	12:49:16	INSERT INTO second_table VALUES (2,'Hasini'), (3,'Rohan')
81	12:49:16	SELECT * FROM first_table UNION ALL SELECT * FROM second_table
82	12:49:16	SELECT name FROM first_table UNION SELECT name FROM second_table
83	12:49:16	SELECT * FROM first_table UNION ALL SELECT * FROM second_table
84	12:49:16	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM se

